

# Markdown to PDF Converter (v1.0.0)

A simple CLI tool built in Go that converts Markdown files into PDF using Pandoc and XeLaTeX.

Built while learning Go — focused on real-world tooling, system interaction, and understanding how tools work under the hood.

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## Version

Current version: **v1.0.0**

This is the first stable CLI release.

This version focuses on correctness, Unicode handling, and reliable PDF generation.

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## Features

- Convert `.md` files to `.pdf`
  - Uses `xelatex` for full Unicode support
  - Custom margin configuration
  - Handles Pandoc execution errors properly
  - Clean and minimal CLI interface
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## Requirements

Make sure the following are installed:

- Go (1.20+ recommended)
  - Pandoc
  - MiKTeX (or any LaTeX distribution)
  - XeLaTeX engine enabled
  - A Unicode font (recommended: Noto Sans)
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## Installation

Clone the repository:

```
git clone https://github.com/SurajTechsmith/markdown-to-pdf
cd markdown-to-pdf
```

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## Usage

Run directly:

```
go run main.go -in file.md
```

Or build an executable:

```
go build -o md2pdf  
./md2pdf -in file.md
```

The tool internally executes:

```
pandoc input.md -o output.pdf --pdf-engine=xelatex -V geometry:margin=1in
```

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## What This Project Helped Me Learn

- Executing external commands using Go (`exec.Command`)
  - Understanding how Pandoc converts Markdown  $\rightarrow$  LaTeX  $\rightarrow$  PDF
  - Unicode handling in XeLaTeX
  - Debugging CLI exit errors
  - Font configuration issues in MiKTeX
  - Structuring a small but complete CLI tool
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## Future Goals (Roadmap)

### v1.x Improvements

- Add custom margin flag from CLI
- Improve error output formatting
- Add verbose/debug mode
- Add unit tests for argument parsing

### v2.0 – Desktop Application

Convert this CLI tool into a cross-platform desktop application using Fyne:

- File picker for selecting Markdown files
- Output location selector
- Convert button with progress feedback
- Status/log panel
- Font selector option
- Drag & drop support
- Cross-platform builds (Windows / Linux / macOS)

### **Long-Term Vision**

- Batch file conversion
  - Configuration file support
  - Predefined themes/templates
  - GitHub release binaries
  - Possibly turn it into a lightweight Markdown publishing tool
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### **Author**

Suraj Yadav Learning Go by building real tools